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ASSESSMENT 1

CSA1006-SOFTWARE ENGINEERING

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Problem: Lean + XP in a Real-Time Tutoring Platform

Scenario

A team is developing a real-time tutoring platform where students can attend live classes, ask questions, and receive instant feedback from tutors. The team wants to reduce waste using Lean principles and improve software quality using Extreme Programming (XP) to deliver reliable and high-quality software quickly.

1. List of Lean Wastes

The following wastes may occur during software development:

- Partially Done Work
 - Features are started but not completed, causing delays.
- Extra Features
 - Developing unnecessary features that users do not need.
- Waiting
 - Developers wait for approvals, feedback, or testing before continuing work.
- Task Switching
 - Frequently changing between different tasks reduces productivity.
- Defects
 - Software bugs require additional time for fixing and testing.

Example

If developers spend time creating advanced chat features before completing the basic live tutoring feature, it creates unnecessary waste.

2. Explanation of Chosen XP Practices

The team should use the following Extreme Programming (XP) practices:

a) Pair Programming

- Two developers work together on the same code.
- This improves code quality and reduces errors.

b) Test-Driven Development (TDD)

- Write test cases before writing the actual code.
- Ensures the software works correctly from the beginning.

c) Continuous Integration

- Merge code changes regularly.
- Automatically test the application after every update.

d) Simple Design

- Keep the software design simple and easy to understand.

e) Refactoring

- Improve the internal code structure without changing its functionality.

3. User Story + Acceptance Criteria

User Story

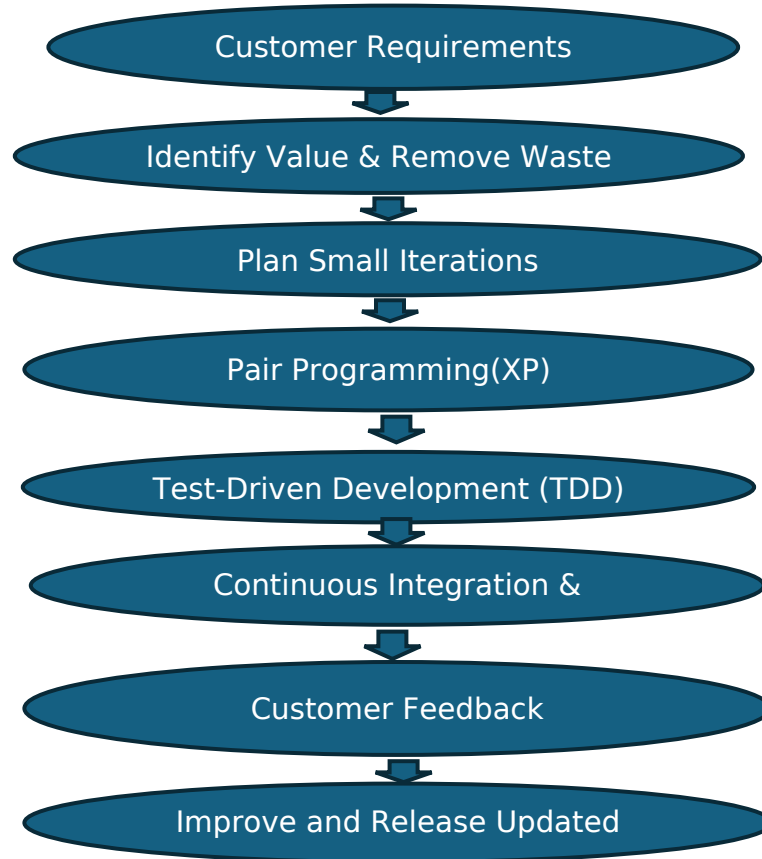
As a student, I want to join a live tutoring session and ask questions in real time so that I can understand the lesson better and receive immediate support from the tutor.

Acceptance Criteria

- The student can join a live tutoring session successfully.
- The student can send questions during the session.
- The tutor can respond to questions instantly.
- The system should notify users if the internet connection is lost.
- The platform should support multiple students without performance issues.

4. Lean + XP Workflow Description

The development workflow combines Lean and XP practices:



Real-World Example

An online tutoring platform allows students to join live classes with teachers.

To improve the system, the development team:

- Removes unnecessary features before release.
- Uses Pair Programming to reduce coding mistakes.

- Applies Test-Driven Development to improve software quality.
- Integrates and tests new code automatically.
- Collects user feedback after every release.

This helps deliver reliable software faster with fewer bugs and better user satisfaction.

Conclusion

Using Lean principles helps reduce unnecessary work, save time, and improve productivity. Extreme Programming (XP) practices such as Pair Programming, Test-Driven Development, Continuous Integration, and Refactoring improve software quality and reliability. By combining Lean and XP, the real-time tutoring platform can deliver high-quality software quickly while meeting customer needs effectively.